Internet dangers - types of attacks, threats to children and defense strategies

1. What are internet dangers?

Definition:

Internet dangers refer to **any risks associated with using the internet**, including malicious software, scams, harmful content, and contact with dangerous individuals.

Why they matter?

- They can harm someone's **privacy**, **finances**, **mental health** or even **physical safety**.
- Children and teenagers are often **more vulnerable**.

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2. Forms and types of cyberattacks

Cyberattacks come in many **forms** (how they are carried out) and **types** (what technique or goal they use). A clear way to understand them is to group them by method and purpose.

1. Malware-based attacks

Attacks that use malicious software to damage, spy on, or take control of systems.

- **Viruses** – attach to legitimate files and spread when those files are executed.
- **Worms** – self-replicate and spread automatically across networks.
- **Trojans** – appear legitimate but secretly perform malicious actions.
- **Ransomware** – encrypts data and demands payment to restore access.
- **Spyware / keyloggers** – secretly monitor user activity, and steal data. A keylogger (short for *keystroke logger*) is a type of software or hardware that records what a user types on a keyboard.

2. Social engineering attacks

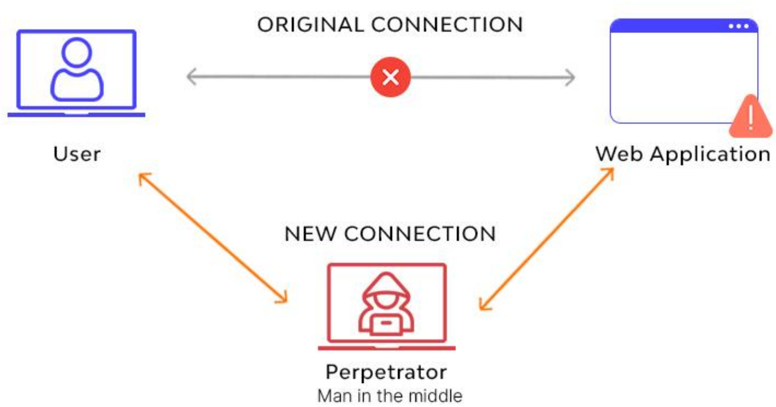
Attacks that exploit human behavior rather than technical vulnerabilities.

- **Phishing** – fraudulent emails or messages that trick users into revealing credentials or clicking malicious links.
- **Spear phishing** – targeted phishing aimed at specific individuals or organizations.
- **Whaling** – phishing attacks targeting senior executives.
- **Pretexting** – attacker invents a believable scenario to gain information.
- **Baiting** – offering something attractive (e.g., free software) that contains malware.

3. Network-based attacks

Attacks targeting network infrastructure and communication.

- **Denial of Service (DoS)** – overwhelms a system to make it unavailable. Attackers flood the target (server) with so many requests (e.g., connection requests - SYN (synchronize) packets) that the server can't process legitimate ones.
- **Distributed DoS (DDoS)** – DoS attack launched from many compromised systems simultaneously.
- **Man-in-the-Middle (MitM)** – attacker intercepts and possibly alters communication between two parties. This type of attack is typical for a wireless (Wi-Fi) environment.



- **Packet sniffing** – capturing network traffic to steal information.

- **ARP (Address Resolution Protocol) spoofing** – redirecting network traffic by falsifying address mappings. Generally, the aim is to associate the attacker's MAC address (physical address of the network card) with the IP (logical) address of another host, such as the default gateway, causing any traffic meant for that IP address to be sent to the attacker instead.

4. Application-level attacks

Attacks that exploit vulnerabilities in software or web applications.

- **SQL Injection** – inserting malicious SQL code to access or manipulate databases.
- **Cross-Site Scripting (XSS)** – injecting malicious scripts into trusted websites.
- **Cross-Site Request Forgery (CSRF)** – forcing a user to perform unwanted actions while authenticated.
- **Remote Code Execution (RCE)** – running attacker-controlled code on a system.

5. Authentication and credential attacks

Attacks aimed at gaining unauthorized access.

- **Brute-force attacks** – trying many password combinations until one works.
- **Dictionary attacks** – using lists of common passwords.
- **Credential stuffing** – reusing stolen username/password pairs across multiple services.
- **Password spraying** – trying one common password against many accounts.

6. Insider attacks

Threats originating from within the organization.

- **Malicious insiders** – employees intentionally stealing or damaging data.
- **Negligent insiders** – users who unintentionally cause breaches through poor security practices.

7. Advanced and persistent attacks

Highly sophisticated, long-term attacks.

- **Advanced Persistent Threats (APT)** – prolonged, targeted attacks often aimed at governments or large organizations.
- **Zero-day attacks** – exploiting vulnerabilities that are unknown to the vendor and have no security patch yet.

8. Physical and hybrid attacks

Combine cyber and physical elements.

- **USB-based attacks** – infected removable media introduced into systems.
- **Hardware tampering** – inserting malicious components into devices.

Cyberattacks can be directed at individuals, companies, or systems. Common forms include:

Attack Type	Description
Phishing	Fake messages (emails, direct messages (DMs)) to trick users into revealing private data
Malware	Malicious software (viruses, trojans, spyware) that damages or spies
Ransomware	Malware that locks someone's files and demands payment to unlock them
DDoS (Distributed Denial of Service)	Overwhelms a website/server with traffic from multiple sources to make it crash
Man-in-the-Middle (MitM)	Hacker intercepts communication between two users
Social Engineering	Manipulating people to reveal confidential data

Attack Type	Description
Cyberbullying	Harassment or threats via messages, social media, games, etc.

Example scenario:

Someone get an email from a delivery service asking for his credit card to "re-ship" a package. This is **phishing**.

Attacks on children and defense strategies

1. The most common mistakes that children make when using the internet put themselves at risk:

- They publish their own or others' personal information.
- They irresponsibly share their passwords.
- They post attention-grabbing images of themselves and others.
- They upload/download illegal content.
- They believe that their activities on the Internet are anonymous.
- They open spam.
- They forward false information without verifying its authenticity and timeliness.
- They irresponsibly click on any link without knowing whether it leads to a safe site.

2. Internet threats specifically targeting children

Children and teenagers are especially exposed to the following threats:

Online grooming

- An adult builds a relationship with a child to gain trust and exploit them (often sexual abuse or trafficking)
- Often occurs through chats, games, or social media

Cyberbullying

- Bullying using technology (insults, threats, shaming)

- Can happen via messaging apps, social media, forums, games
- Often anonymous

Inappropriate content

- Pornography, violence, drug-related, hate speech
- Can appear in ads, pop-ups, or be shared directly

Scams and fraud

- Fake giveaways, free skins/items in games, “click here to win” messages
- Can lead to account theft or malware installation

Addiction and oversharing

- Spending too much time online
- Sharing private info like name, address, school

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3. Defense strategies

General safety tips:

- Use **strong passwords**, don’t reuse them
- Enable **two-factor authentication**
- Keep software and antivirus updated
- Don’t click suspicious links or download unknown files
- Think before sharing personal information

For children and teens:

- **Don’t talk to strangers online** (even in games)
- Never meet online-only “friends” in real life without parent’s approval
- Tell a **trusted adult** if something seems wrong or uncomfortable
- **Use privacy settings** on social media
- Block and report **cyberbullies**

For parents and educators:

- Talk openly about internet use
- Use **parental controls**
- Monitor app/game activity
- Set screen time limits
- Encourage open communication

Key terms summary

Term	Definition
Phishing	Tricking users into revealing information through fake emails or messages
Malware	Harmful software designed to damage or spy on a system
Grooming	Online manipulation of children by adults for exploitation
Cyberbullying	Harassment using digital tools (messages, posts, comments)
Two-factor authentication	Login that requires password + another verification (e.g. code)
Parental controls	Tools that limit or monitor children's digital activity